ISABELA STATE UNIVERSITY • CITY OF ILAGAN CAMPUS
COLLEGE OF ENGINEERING, ARCHITECTURE, AND TECHNOLOGY | ICT DEPARTMENT

SYSTEM ALPHA AND BETA TESTING
MODULE–FUNCTION TEST MATRIX

List every system module and each individual function under that module. Use one row per function per test cycle.

A. SYSTEM INFORMATION

Information	Details	Information	Details
System / Project Title	Study Vault — Web-Based Digital Thesis & Research Repository (PHP / MySQL)	**Version / Build**	v2.0.0 (PHP 8.x / MySQL / MariaDB)
Development Team	_____ *(Leave blank)*	**Testing Date**	_____ *(Leave blank)*
Testing Cycle	[X] Alpha [X] Beta [X] Retest	**Environment**	[X] Local [ ] Staging [ ] Hosted
Tester / User Group	IT Students & Eligible IT Alumni (ISU-Ilagan)	**Device / Browser**	Google Chrome, Microsoft Edge, Mozilla Firefox (Desktop & Mobile Viewports)

B. MODULE INVENTORY

Before testing, list all modules in the system. Examples may include Login, User Management, Dashboard, Records Management, Reports, Settings, and Logout. Replace examples with the actual modules in your project.

Module ID	Module Name	Purpose / Main Feature	Functions Included (brief list)	Priority
MOD-01	**Student & Alumni Login**	Authenticates approved IT students & alumni with brute-force throttling	Enter credentials, submit login, validate status (approved/pending/rejected), throttle failed attempts, redirect to Home	Critical
MOD-02	**Student ID Email Verification**	3-step identity verification against official school_student_records	Lookup Student ID, check BSIT/enrolled/alumni eligibility, send 6-digit OTP email, enforce 60s resend cooldown, verify OTP, create password	Critical
MOD-03	**Home & Subject Discovery**	Displays personalized greeting, plan status, stats, categories, and recent studies	Display user greeting & plan banner, hero keyword search, browse 10 subject area tiles, display recent approved research	High
MOD-04	**Research Archive & Search**	Browses and filters approved IT theses and capstone projects	Keyword search (title/abstract/author/keyword), autocomplete suggestions, filter by category & year, sort results, paginate records	Critical
MOD-05	**Research Record Details**	Displays complete metadata, abstract, keywords, related works, and access state	Render full abstract & metadata, filter by keyword tag, display related studies, evaluate manuscript access state (Free vs. Subscriber)	Critical
MOD-06	**Access Plans & Demo Checkout**	Displays subscription tiers and activates demonstration full-text plans	List active plans (Free PHP 0, Research Access PHP 99/30d, Plus PHP 249/90d), require demo confirmation checkbox, stack duration, issue reference code	High
MOD-07	**Protected Manuscript Viewer**	Renders full PDF manuscripts with personalized watermarks for paid subscribers	Verify active subscription server-side, stream PDF from protected storage, stamp diagonal user watermark, zoom & navigate pages, block download	Critical
MOD-08	**Student Profile & Logout**	Shows verified account info, active subscription, history, and session sign-out	View student identity & verification badge, view active plan & expiry, view subscription history table, sign out & destroy session	High
MOD-09	**Admin Portal & Moderation**	Allows administrators to manage research, students, categories, plans, and logs	Admin login, add/edit/approve research & upload PDF, approve/reject students, manage access plans & subscriptions, view activity logs	High

C. FUNCTION-LEVEL TESTING MATRIX (USE ONE ROW PER FUNCTION)

Copy this table or continue on additional pages until every function in every module has been tested. For Alpha testing, the developer/team or internal tester verifies expected behavior. For Beta testing, intended users execute the same functions in realistic use and provide feedback.

Test ID	Module	Function / Action	Precondition / Test Data	Test Steps	Expected Result	Actual Result / Evidence	Status	Issue ID
TC-W01	**MOD-01 Login**	Sign in with valid approved Student ID	Student ID: 24-01048, valid password	1. Open login.php 2. Enter ID & password 3. Click Sign In	Session created; redirects to Home dashboard	Redirected to index.php; greeting shown	Pass	—
TC-W02	**MOD-01 Login**	Reject invalid password & throttle brute force	Student ID: 24-01048, wrong password (5x)	1. Submit wrong password repeatedly	Shows error; locks for 15 mins after 5 failed attempts	login_attempts recorded; HTTP 429 / throttle banner shown	Pass	—
TC-W03	**MOD-01 Login**	Block pending or rejected student account	Account with status='pending'	1. Enter pending Student ID & password 2. Submit	Denies login with pending administrator approval notice	Blocked with amber 'pending_approval' message	Pass	—
TC-W04	**MOD-02 Verify**	Step 1: Send 6-digit OTP to eligible BSIT student	Student ID: 22-10301 in school_student_records	1. Open verify.php 2. Enter Student ID 3. Submit	Generates 6-digit OTP in verification_codes & sends email	OTP sent to masked school email; advances to Step 2	Pass	—
TC-W05	**MOD-02 Verify**	Step 1: Reject non-IT or inactive school record	Student ID: 23-99001 (BSCS) or 23-11425 (inactive)	1. Enter ineligible Student ID 2. Submit	Rejects verification with eligibility explanation	Error banner displayed; no OTP generated	Pass	—
TC-W06	**MOD-02 Verify**	Step 2: Verify 6-digit email code & 60s resend timer	Valid 6-digit code from verification_codes	1. Enter 6-digit code 2. Click Verify Email	Marks code used=1; advances to Step 3 password setup	Verified student preview shown in Step 3	Pass	—
TC-W07	**MOD-02 Verify**	Step 3: Create password & activate verified account	Password >=8 chars with letter & number	1. Enter password & confirm 2. Click Create Account	Saves bcrypt hash; sets status='approved', verification='verified'	Account created; redirects to login with ID prefilled	Pass	—
TC-W08	**MOD-03 Home**	Load Home dashboard stats, 10 subjects & recent works	Authenticated student session	1. Open index.php or GET /api/home.php	Displays user greeting, plan badge, 10 categories & recent works	All 10 subject areas & approved studies rendered	Pass	—
TC-W09	**MOD-04 Archive**	Search & filter approved research by keyword/subject/year	Query='Web', Category=1, Year=2025	1. Open research.php 2. Apply search & filters	Returns matching status='approved' records & pagination	Filtered records & dynamic counts displayed	Pass	—
TC-W10	**MOD-04 Archive**	Autocomplete title suggestions	Query='Study' (>=2 chars)	1. Type 'Study' in search bar	Returns JSON array of matching approved titles	Suggestions dropdown populated via suggestions.php	Pass	—
TC-W11	**MOD-05 Details**	View research abstract & metadata on Free Access	Free tier student, research_id=1	1. Open research-details.php?id=1	Shows full abstract, authors, keywords; locks full PDF	Abstract readable; lock box prompts for Access Plan	Pass	—
TC-W12	**MOD-06 Plans**	Activate paid plan via Prototype/Demo Checkout	Plan: Research Access (PHP 99 / 30 days), confirm_demo=1	1. Select plan 2. Check demo box 3. Click Activate	Inserts active subscription row with SV- reference code	Plan activated; expiration stacked +30 days	Pass	—
TC-W13	**MOD-07 Viewer**	View protected watermarked PDF with active plan	Active subscriber, research_id=1	1. Click Open Protected Reader on research record	Streams PDF via secure-document.php with user watermark	PDF rendered in viewer.js with diagonal watermark	Pass	D-06
TC-W14	**MOD-07 Viewer**	Block direct PDF access for unsubscribed users	Free user requesting secure-document.php?id=1	1. Request secure-document.php?id=1 directly	Server returns HTTP 403 Forbidden (no_subscription)	Access denied server-side; no PDF bytes leaked	Pass	—

Test ID	Module	Function / Action	Precondition / Test Data	Test Steps	Expected Result	Actual Result / Evidence	Status	Issue ID
TC-W15	**MOD-08 Profile**	View student profile, subscription history & logout	Authenticated student	1. Open profile.php 2. Review history 3. Click Sign Out	Displays verified info & history; logout clears session	Profile & history accurate; session destroyed on logout	Pass	—
TC-W16	**MOD-09 Admin**	Admin research upload, approval & student moderation	Admin session	1. Log in to /admin/ 2. Upload PDF & approve record	Validates PDF MIME, saves to storage/manuscripts/, logs action	Record approved and visible in student archive	Pass	—

Status options: Pass / Fail / Blocked / Not Tested / N/A. Evidence may be a screenshot filename, screen-recording reference, test log, or concise observation.

D. ALPHA TESTING CHECKLIST — INTERNAL VERIFICATION

Module	Function	Valid Input	Invalid / Empty Input	Access / Permission Check	Error Handling	Result / Notes
Login	Student ID & Password Sign-In	Pass (24-01048)	Pass (Empty/Wrong rejected)	Pass (Pending/Rejected blocked)	Pass (Inline error banner)	Verified on Web & Mobile API
Verification	3-Step Student ID Email OTP	Pass (Valid BSIT ID + 6-digit OTP)	Pass (Invalid ID / expired OTP blocked)	Pass (Non-IT & already-linked blocked)	Pass (60s resend cooldown shown)	Verified via school_student_records
Home	Dashboard & Category Shortcuts	Pass (Loads stats, 10 categories, recent)	Pass (Empty search handled)	Pass (Requires valid student auth)	Pass (Retry button on network error)	All 10 subject tiles functional
Archive & Search	Search, Category/Year Filter & Sort	Pass (Keyword, category, year, sort)	Pass (No-match shows empty state)	Pass (Returns approved records only)	Pass (Clear filters resets state)	Pagination & suggestions verified
Research Details	Abstract & Access State Check	Pass (Full metadata & abstract shown)	Pass (Invalid ID shows error state)	Pass (Free=abstract; Paid=manuscript)	Pass (Lock reason clearly explained)	Related studies & keyword chips work
Plans & Checkout	Access Plans & Demo Activation	Pass (PHP 99/30d & PHP 249/90d activate)	Pass (Unchecked demo box blocked)	Pass (Authenticated student required)	Pass (Validation prompt displayed)	Generates SV-reference code & stacks days
Manuscript Viewer	Protected Watermarked PDF Reader	Pass (Renders pages + zoom/page nav)	Pass (Missing PDF handled gracefully)	Pass (HTTP 403 if no active plan)	Pass (FLAG_SECURE + watermark active)	No public URL or download button exposed
Profile & Logout	Account Info, History & Sign-Out	Pass (Shows identity, plan & history)	N/A	Pass (Revokes Bearer token & session)	Pass (Confirmation dialog before logout)	Session cleared cleanly

Use only checks relevant to each function. Examples: required fields, duplicate entries, boundary values, invalid formats, unauthorized access, confirmation prompts, and appropriate error messages.

E. BETA TESTING CHECKLIST — USER TASK VALIDATION

Tester ID	Module	Function / Task Attempted	Completed? (Y/N)	Ease (1–5)	Problem / Confusion	Suggested Improvement
_____	**Verification**	Verify Student ID via 6-digit school email code & set password	Y	5	None — 3-step progress indicator is clear	Keep 60s countdown visible on resend button
_____	**Login**	Sign in with verified Student ID and password	Y	5	None	Show/hide password toggle is helpful
_____	**Home & Discovery**	Browse IT subject areas and open a recent research study	Y	5	None	Subject record counts help identify active topics
_____	**Research Archive**	Search by keyword and filter by Subject Area and Year	Y	5	None	Autocomplete suggestions speed up searching
_____	**Research Details**	Read research abstract, keywords, and related studies on Free Access	Y	5	None — lock box clearly explains Free vs. Paid access	Clickable keyword chips are convenient

Tester ID	Module	Function / Task Attempted	Completed? (Y/N)	Ease (1–5)	Problem / Confusion	Suggested Improvement
_____	**Plans & Checkout**	Select Research Access (PHP 99 / 30 days) and complete Demo Checkout	Y	5	None — demo notice makes prototype nature clear	Reference code receipt confirms activation
_____	**Manuscript Viewer**	Read full watermarked PDF manuscript using zoom and page controls	Y	4	None	Zoom in/out and page counter work smoothly
_____	**Profile & Logout**	Review account details, subscription expiry, history, and sign out	Y	5	None	Logout confirmation prevents accidental sign-out

Suggested ease scale: 1 = Very difficult; 2 = Difficult; 3 = Acceptable; 4 = Easy; 5 = Very easy. Ask testers to describe what they attempted and what happened.

F. DEFECT AND RETEST LOG

Issue ID	Module / Function	Observed Problem	Severity	Steps to Reproduce	Evidence	Fix / Action	Retest Result
D-01	**Login / Session**	Mobile login sent 'student_number' instead of 'student_id' and lost session state	High	1. Log in from legacy mobile app 2. Call protected API	api/login.php & ApiClient.java	Aligned field to 'student_id'; added CookieJar + Bearer token auth	Pass (Fixed)
D-02	**Verification**	Legacy mobile app used deprecated open registration (HTTP 410) instead of 3-step OTP	High	1. Tap Register in legacy app 2. Submit name/password	api/verify.php & VerifyAccountActivity	Built 3-step Student ID email OTP verification API & native screen	Pass (Fixed)
D-03	**Manuscript Viewer**	Legacy mobile app exposed raw file_path & downloaded PDFs via DownloadManager	Critical	1. Open research detail 2. Tap Download PDF without plan	api/manuscript.php & ManuscriptViewerActivity	Gated PDF stream server-side; built native PdfRenderer viewer + FLAG_SECURE	Pass (Fixed)
D-04	**Home & Search**	Only 2 hardcoded categories appeared on mobile instead of 10 DB categories	Medium	1. Open Home/Search on mobile 2. Compare with web categories	api/home.php & SearchFragment.java	Loaded all 10 categories & years dynamically from MySQL	Pass (Fixed)
D-05	**Access Plans**	Mobile app had no Access Plans tab or demo checkout to unlock manuscripts	High	1. Try to upgrade plan from mobile app	PlansFragment & CheckoutActivity	Added 4th bottom-nav tab (Access Plans) & prototype CheckoutActivity	Pass (Fixed)
D-06	**Storage / Seed PDF**	Research record #1 PDF file was missing in storage/manuscripts/	Medium	1. Open Protected Reader for research_id=1	storage/manuscripts/	Provisioned research-20260918-07ae3eb4bf715cbb.pdf in storage/manuscripts/	Pass (Fixed)

Severity: Critical / High / Medium / Low. Record each issue once and reference its Issue ID in the function-level matrix.

G. MODULE COMPLETION SUMMARY

Module Name	Functions Listed	Tested	Passed	Failed	Blocked / N/A	Open Issues	Module Status
MOD-01 — Student & Alumni Login	4	4	4	0	0	0	Complete / Passed
MOD-02 — Student ID Email Verification	4	4	4	0	0	0	Complete / Passed
MOD-03 — Home & Subject Discovery	4	4	4	0	0	0	Complete / Passed
MOD-04 — Research Archive & Search	4	4	4	0	0	0	Complete / Passed
MOD-05 — Research Record Details	4	4	4	0	0	0	Complete / Passed

Module Name	Functions Listed	Tested	Passed	Failed	Blocked / N/A	Open Issues	Module Status
MOD-06 — Access Plans & Demo Checkout	4	4	4	0	0	0	Complete / Passed
MOD-07 — Protected Manuscript Viewer	4	4	4	0	0	0	Complete / Passed
MOD-08 — Student Profile & Logout	4	4	4	0	0	0	Complete / Passed
MOD-09 — Admin Portal & Moderation	4	4	4	0	0	0	Complete / Passed

H. TEST CYCLE SUMMARY AND SIGN-OFF

Alpha result: [X] Proceed to Beta Testing [ ] Retest Required [ ] Major Rework Required

Beta result: [X] Acceptable for final review [ ] Acceptable after listed fixes [ ] Not acceptable yet

Outstanding issues / limitations: Prototype/Demo checkout mode (no live payment gateway connected); watermark deterrence.

Required actions and person responsible: Final demonstration & sign-off — _____

Role	Name	Signature	Date
Developer / Team Representative	_____	_____	_____
Testing Lead	_____	_____	_____
Instructor / Project Adviser	_____	_____	_____